# Appendix

GranMix: Adaptive Partition Granularity Management
for Online Data Mixing

*Reading note.* This document contains the four appendix sections from the current manuscript. References to numbered main-paper results, algorithms, sections, and unlettered equations retain the manuscript's numbering. Appendix equations and the appendix proposition use lettered numbering. The bibliography includes only works cited here.

## A Local Validation Gains and Partition Regret

### A.1 One-step approximation

Fix a pre-update state and domain probe. Taylor's theorem applied along $\theta_s - h_s g_s(i)$ gives the local validation-gain expansion in Section 3.1. Its remainder obeys $|R_s(i)| \le L_{d(i)} h_s^2 \|g_s(i)\|^2/2$. Dividing by $h_s a^V_{s,d(i)}$ and using that $\mathrm{clip}_M$ is 1-Lipschitz yields

$$\left| \mathrm{clip}_M\left( \frac{\Delta_s(i)}{h_s a^V_{s,d(i)}} \right) - u_s(i) \right| \le \min\left\{ 2M, \frac{L_{d(i)} h_s \|g_s(i)\|^2}{2 a^V_{s,d(i)}} \right\}. \tag{A.1}$$

This is the first-order local contribution approximation used in the controller. The normalization is evaluated at the pre-update state and held fixed in the local expansion; each domain has its own validation objective and scale. The scoring update is a gradient-descent step, as in the first-order construction of GREATS [1].

Let $v_t^{\mathrm{loc}}(i)$ average the clipped normalized one-step gains on the left of Equation (A.1), and let $\rho_t(i)$ average its right-hand side over $\mathcal{W}_t$. Then $|v_t^{\mathrm{loc}}(i) - v_t(i)| \le \rho_t(i)$. Replacing $v_t$ by $v_t^{\mathrm{loc}}$ in Equation (1) defines $J_t^{\mathrm{loc}}$, for which

$$\begin{aligned} J_t^{\mathrm{loc}}(q) - J_t^{\mathrm{loc}}(q_t^-) \ge\ & J_t(q) - J_t(q_t^-) \\ & - \sum_i (q(i) + q_t^-(i)) \rho_t(i). \end{aligned} \tag{A.2}$$

The same local Taylor remainder therefore transfers the window control comparison to normalized finite-step validation gains.

### A.2 Proof of Theorem 1

Fix a boundary and write $q^- = q_t^-$ and $v = v_t$ for this proof. Define

$$Z_{\mathrm{sample}} = \sum_i q^-(i) e^{\eta v(i)}, \qquad q^\star(i) = \frac{q^-(i) e^{\eta v(i)}}{Z_{\mathrm{sample}}}. \tag{A.3}$$

Direct substitution gives, for any sample distribution $q$,

$$J_t(q) = \frac{1}{\eta} \log Z_{\mathrm{sample}} - \frac{1}{\eta} D_{\mathrm{KL}}(q \| q^\star). \tag{A.4}$$

Thus $J_t^\star = \eta^{-1} \log Z_{\mathrm{sample}}$, and $Z_{\mathrm{sample}} = \mathbb{E}_p[e^{\phi_t(i)}]$.

For a fixed cut, expand the KL using $q_{\Pi,w}(i) = w_G p(i \mid G)$:

$$D_{\mathrm{KL}}(q_{\Pi,w} \| q^-) = \sum_{G \in \Pi} w_G \log \frac{w_G}{p(G)} - \sum_{G \in \Pi} w_G \kappa_t(G). \tag{A.5}$$

Let $a_G = p(G)e^{\eta v_t(G)+\kappa_t(G)}$ and $Z_\Pi = \sum_G a_G$. Then

$$J_t(q_{\Pi,w}) = \frac{1}{\eta}\log Z_\Pi - \frac{1}{\eta}D_{\mathrm{KL}}(w\|\{a_G/Z_\Pi\}_G)\,. \tag{A.6}$$

Consequently $J_t^{\Pi,\star} = \eta^{-1}\log Z_\Pi$; subtracting the optimal values proves Equation (3). For each group, Jensen's inequality yields

$$\mathbb{E}_{p(\cdot|G)}[e^{\phi_t(i)}] \geq e^{\bar{\phi}_t(G)}.$$

Averaging with positive masses $p(G)$ proves nonnegativity. Strict convexity of the exponential gives equality exactly when $\phi_t$ is constant within every group.

For the expansion, take $q^- = p$. Since $|v(i)| \leq M$, the Taylor expansions of both log moment-generating functions have uniformly bounded third-order remainders near $\eta = 0$. Their linear terms coincide. Their quadratic coefficients differ by

$$\begin{aligned}&\mathrm{Var}_p[v(i)] - \mathrm{Var}_{G\sim p(G)}[v_t(G)]\\ &\quad= \sum_{G\in\Pi} p(G)\,\mathrm{Var}_{p(\cdot|G)}[v(i)],\end{aligned} \tag{A.7}$$

by the law of total variance. Let $V_t^{\mathrm{within}}(\Pi)$ denote this common value. Dividing the log-value difference by $\eta$ gives

$$\mathcal{R}_t(\Pi) = \frac{\eta}{2}V_t^{\mathrm{within}}(\Pi) + O(\eta^2). \tag{A.8}$$

### A.3 Separable regularization

Fix a boundary and suppress its time index. Let $f$ be lower semicontinuous and strictly convex on its effective domain in $[0,\infty)$, and continuously differentiable on $(0,\infty)$. Consider

$$J_f(q) = \sum_i q(i)v(i) - \eta^{-1}\sum_i q^-(i) f\!\left(\frac{q(i)}{q^-(i)}\right). \tag{A.9}$$

Assume its sample-level maximum is attained at a unique interior distribution $q_f^\star$. This class uses the same cost function for every relative intervention; KL corresponds to $f(x) = x\log x - x + 1$. Groups continue to use $p(i \mid G)$, so the relative intervention against $q^-$ and the multiplier against $p$ need not coincide.

**Proposition A.1** (Lossless grouping with separable costs)**.** *A partition has zero loss under $J_f$ exactly when $q_f^\star(i)/p(i)$ is constant inside each group. When $q^-/p$ is constant inside every group, this is equivalent to equal contributions $v(i)$ within each group; the common-reference case $q^- = p$ is a special case.*

*Proof.* Write $x_i = q(i)/q^-(i)$, $x_i^\star = q_f^\star(i)/q^-(i)$, and $D_f(a,b) = f(a) - f(b) - f'(b)(a-b)$. Interior optimality gives $f'(x_i^\star) = \eta(v(i) - \lambda)$ for a common normalization multiplier $\lambda$. For any finite-value sampler, normalization cancels the linear term in $\lambda$, yielding

$$J_f(q_f^\star) - J_f(q) = \eta^{-1}\sum_i q^-(i) D_f(x_i, x_i^\star). \tag{A.10}$$

The partition loss is the minimum of this expression over $q = q_{\Pi,w}$. Strict convexity makes each term nonnegative and gives equality only when $q = q_f^\star$. The group-weight simplex is compact, and lower semicontinuity ensures attainment. Hence the minimum is zero exactly when the unrestricted optimum is representable as $w_G p(i \mid G)$, or equivalently when $q_f^\star(i)/p(i)$ is groupwise constant.

If $q^-(i)/p(i)$ is constant within a group, groupwise equality of $q_f^\star(i)/p(i)$ is equivalent to equality of $x_i^\star$. Since $f'$ is strictly increasing, the first-order condition makes this equivalent to equality of $v(i)$. For KL, the optimum is the exponential tilt and yields the control profile in Theorem 1. Its exponential form also produces the additive scores used in Section 4. □

# B Diagnostic Statistics and Online Calibration

## B.1 History statistics

Use block length $\tau$ dividing $H$, so window boundaries coincide with block boundaries. Let $\mathcal{B}_t$ contain up to the latest $K_{\text{hist}} \geq H/\tau$ completed blocks at boundary $t$, and let $b_\star$ be the most recent one. Completed-window blocks are retained until their audit is finished. Set $h_{t,b} = \lambda_{\text{hist}}^{b_\star - b}$, where $0 < \lambda_{\text{hist}} \leq 1$. For each node, compute

$$\begin{aligned} W_t(G) &= \sum_{b \in \mathcal{B}_t} h_{t,b} W_{b,G}, \\ W_{2,t}(G) &= \sum_{b \in \mathcal{B}_t} h_{t,b}^2 W_{2,b,G}, \\ T_{1,t}(G) &= \sum_{b \in \mathcal{B}_t} h_{t,b} S_{b,G}/a_b, \\ T_{2,t}(G) &= \sum_{b \in \mathcal{B}_t} h_{t,b} Q_{b,G}/a_b^2. \end{aligned} \tag{B.1}$$

When $W_t(G) > 0$, the normalized mean, variance, and effective count are

$$\begin{aligned} \bar{z}_t(G) &= T_{1,t}(G)/W_t(G), \\ s_t^2(G) &= \max\{0, T_{2,t}(G)/W_t(G) - \bar{z}_t(G)^2\}, \\ n_{\text{eff},t}(G) &= W_t(G)^2/W_{2,t}(G). \end{aligned} \tag{B.2}$$

A node with zero total weight has effective count zero and is invalid. The square on $h_{t,b}$ in $W_{2,t}$ accounts for the decay applied to each observation's importance weight.

For blocks with $W_{b,G} > 0$, define $\mu_{b,G} = S_{b,G}/(a_b W_{b,G})$ and $\pi_{t,b,G} = h_{t,b} W_{b,G}/W_t(G)$. The block-dispersion feature is

$$s_t^{\text{blk}}(G)^2 = \sum_{\substack{b \in \mathcal{B}_t \\ W_{b,G} > 0}} \pi_{t,b,G} \big(\mu_{b,G} - \bar{z}_t(G)\big)^2. \tag{B.3}$$

Compute $\bar{z}_t^{\text{recent}}(G)$ with the same formulas over the latest $1 \leq K_{\text{recent}} \leq K_{\text{hist}}$ blocks, and set $d_t(G) = |\bar{z}_t^{\text{recent}}(G) - \bar{z}_t(G)|$. If this shorter history is empty for $G$, set $\bar{z}_t^{\text{recent}}(G) = \bar{z}_t(G)$. The mixer signal $m_t(G)$ is $\mu_{b,G}$ from the latest block in $\mathcal{B}_t$ with positive node weight. Finally, $a_t = a_{b_\star}$, with $a_b$ defined in Section 3.2.

These features enter the base-scale formula in Section 3.2. The coefficients are nonnegative and $b_{\min} > 0$. History lengths, decay, and the three nonnegative feature coefficients are fixed before a training run. The width multiplier is updated only by Equation (4). Both $\hat{v}_t(G)$ and $b_t(G)$ are retained in the prediction record until their window is audited.

## B.2 Completed-window audit

The audit estimates the reference-conditional mean over the completed window from the logged draws. Its martingale construction accommodates adaptive sampling, following the general approach used for off-policy inference [2]. Index the $N = HB$ draws in a window by $j$. Before a draw, the model state, domain probes, local contribution function, and sampling probabilities are fixed by the available history. Write $u_{t,j}(i)$ for the corresponding contribution, with $|u_{t,j}(i)| \leq M$. For node $G$, let

$$X_{t,j,G} = \frac{\mathbf{1}\{I_{t,j} \in G\} p(I_{t,j})}{p(G) q_{t,j}(I_{t,j})} \, u_{t,j}(I_{t,j}). \tag{B.4}$$

With $\mathcal{F}_{t,j-1}$ denoting this pre-draw history,

$$\mathbb{E}[X_{t,j,G} \mid \mathcal{F}_{t,j-1}] = \mu_{t,j,G} := \sum_{i \in G} p(i \mid G) u_{t,j}(i). \tag{B.5}$$

Since every optimizer step has $B$ draws, $N^{-1}\sum_j \mu_{t,j,G} = v_t(G)$. Thus $\sum_j (X_{t,j,G} - \mu_{t,j,G})$ is a martingale sum, including when the model depends on earlier training draws.

The coverage floor gives the deterministic bounds

$$\begin{aligned} |X_{t,j,G} - \mu_{t,j,G}| &\le b_G^{\mathrm{aud}} := \frac{2M}{\varepsilon p(G)}, \\ \sum_{j=1}^{N} \mathrm{Var}(X_{t,j,G} \mid \mathcal{F}_{t,j-1}) &\le V_G^{\mathrm{aud}} := \frac{NM^2}{\varepsilon p(G)}. \end{aligned} \tag{B.6}$$

Indeed, the conditional second moment is at most $M^2/[\varepsilon p(G)]$, and $|\mu_{t,j,G}| \le M$. Allocate $\delta_t = \delta_{\mathrm{aud}}/[t(t+1)]$ to window $t$, and let $\chi_t = \log(2|\mathcal{T}|/\delta_t)$. A two-sided Bernstein–Freedman bound [3] gives the untruncated radius

$$R_t^{\mathrm{aud}}(G) = \frac{b_G^{\mathrm{aud}}\chi_t/3 + \sqrt{2V_G^{\mathrm{aud}}\chi_t + (b_G^{\mathrm{aud}}\chi_t/3)^2}}{N}. \tag{B.7}$$

Substituting Equation (B.6) and setting $x_t(G) = \chi_t/(N\varepsilon p(G))$ gives the audit-radius formula in Section 3.2 exactly after truncation at $2M$. The raw estimate $N^{-1}\sum_j X_{t,j,G}$ is $S_{t,G}^{\mathrm{win}}/(Np(G))$; its clipped version and endpoints are the audit-estimate and audit-interval definitions in Section 3.2.

Projection onto $[-M, M]$ cannot increase the estimation error because $v_t(G)$ belongs to this range. When the untruncated radius exceeds $2M$, the interval is the entire range. A union bound over all nodes and windows, using $\sum_{t\ge1}\delta_t = \delta_{\mathrm{aud}}$, establishes

$$\mathbb{P}\Big(\forall t, G:\ v_t(G) \in [\check{L}_t(G), \check{U}_t(G)]\Big) \ge 1 - \delta_{\mathrm{aud}}. \tag{B.8}$$

Only the nodes in the saved $\mathcal{V}_t$ need be evaluated computationally. The guarantee covers the full fixed tree, so the validity filtering and subsequent cut selection preserve it. A node with no draws has raw audit sum zero and is still included in its saved record. Both the audit sum and the sample count are obtained from the same training log; the audit does not require a separate validation trajectory.

### B.3 Proof of Proposition 1

For this proof, let $d_t^{\mathrm{aud}}(G)$ be the larger distance from the saved forecast to the audit endpoints, and set

$$S_t = \max_{G\in\mathcal{V}_t} \frac{d_t^{\mathrm{aud}}(G)}{b_t(G)},$$

with zero for an empty set. The interval-containment test in Section 3.2 is exactly $\mathrm{err}_t = \mathbf{1}\{S_t > c_t\}$.

The saved forecasts and audit endpoints lie in $[-M, M]$, and $b_t(G) \ge b_{\min}$. Therefore $0 \le S_t \le S_{\max} := 2M/b_{\min}$. If $c_t > S_{\max}$, then $\mathrm{err}_t = 0$ and the update cannot increase $c_t$; otherwise it increases by at most $\gamma_{\mathrm{cal}}$. Induction yields $0 \le c_t \le C_{\max}$.

From the lower projection in Equation (4),

$$c_{t+1} - c_t \ge \gamma_{\mathrm{cal}}(\mathrm{err}_t - \zeta). \tag{B.9}$$

Summing and rearranging gives

$$\sum_{t=1}^{T}\mathrm{err}_t \le \zeta T + \frac{c_{T+1} - c_1}{\gamma_{\mathrm{cal}}} \le \zeta T + \frac{C_{\max} - c_1}{\gamma_{\mathrm{cal}}}. \tag{B.10}$$

On the event in Equation (B.8), $S_t \le c_t$ means that every audit interval for $G \in \mathcal{V}_t$ is contained in its prediction interval. Hence $z_t \le \mathrm{err}_t$ on that event, proving Equation (5) simultaneously for every $T$. This argument uses the chronological prediction records and permits arbitrary changes in the local contribution sequence and in the valid node set.

# C Refinement Guarantees

## C.1 Conservative gain and target weights

Fix one boundary and use the time-suppressed notation of Section 4. The definitions of $e(G)$ and $E(\Pi, w)$ in Section 4.1 imply

$$L(G) - e(G) \le v(G) \le U(G) + e(G). \tag{C.1}$$

Average the lower bound with the candidate weights and the upper bound with the current weights. Subtracting the common KL penalty proves Proposition 2. The inequality is pointwise in the contribution vector, so it holds for the data-dependent cut and weights returned by the controller.

For Theorem 2, expand the KL by Equation (A.5) and substitute the node scores. With $Z_\Pi = \sum_{G\in\Pi} A(G)$ and $w^\star_G = A(G)/Z_\Pi$,

$$C(\Pi, w) = \eta^{-1} \log Z_\Pi - U^- - \eta^{-1} D_{\mathrm{KL}}(w \| w^\star). \tag{C.2}$$

Nonnegativity of KL gives the optimizer and optimal value. Positivity of the node scores makes the optimizer unique. Equation (15) is the same identity evaluated at executable weights.

## C.2 Dynamic programming

Every cut of a subtree either contains its root or covers every child subtree with a child cut. For a leaf, selecting the leaf is the only feasible cut, and its table entry is finite exactly when the leaf is valid. Suppose all child tables are correct. Selecting the parent contributes $A(G)$ with one group when valid. Otherwise, any feasible cut allocates positive group counts $k_1, \dots, k_d$ to the children, whose table values are given by the induction hypothesis. Maximizing over these allocations proves the recurrence and its converse: each finite maximizing choice corresponds to a valid cut. Induction to the root and the budget maximization prove optimality.

Combining a child table with the running table is a truncated max-sum convolution of two arrays of length at most $m + 1$, costing $O(m^2)$. There is one operation per tree edge, giving $O(|\mathcal{T}|m^2)$ search time. Tables and backtracking allocations require $O(|\mathcal{T}|m)$ space, including the intermediate choices recorded at each edge. The two handoff calls are additional execution costs.

For numerical stability, all scores may be rescaled by the common factor $e^{-a}$, where $a = \max_{G\in\mathcal{V}} \log A(G)$. This preserves every maximizing cut and all normalized weights. The target gain recovered from a rescaled root value uses $\eta^{-1}(\log F(r, k) + a) - U^-$. The same offset is used throughout the tree.

# D Handoff and Executed Decisions

## D.1 Proof of Theorem 3

Substituting the inherited reference into Equation (A.5) gives Equation (12), since $\log r^0(G) = \log p(G) + \kappa(G) - \log Z$. Jensen's inequality gives $p(G)e^{\kappa(G)} \le q^-(G)$, so $Z \le 1$. Without coverage floors, the minimum KL on the cut is $-\log Z$, attained at $r^0$.

The floor-constrained simplex is compact and nonempty because the floor masses sum to $\varepsilon < 1$. Write $f(w) = D_{\mathrm{KL}}(w\|\widetilde{w})$ and $g(w) = D_{\mathrm{KL}}(w\|r^0)$; both are strictly convex. For $\alpha \in [0, 1]$, minimizing their convex combination is equivalent to minimizing

$$\sum_G w_G \log \frac{w_G}{z_G}, \qquad z_G = \widetilde{w}_G^{\alpha} r^0(G)^{1-\alpha}.$$

The KKT conditions give

$$w_G(\alpha) = \max\{\ell_G, z_G/Z_\alpha\}.$$

Positivity of $z_G$ and total floor mass below one give a unique positive normalizing denominator, proving Equation (14).

For $\alpha > 0$, the same vector minimizes $f(w) + \lambda g(w)$, with $\lambda = (1-\alpha)/\alpha$. Write $w_\lambda$ for this optimizer. Comparing the optimality inequalities at $\lambda_1 > \lambda_2$ gives

$$(\lambda_1 - \lambda_2)(g(w_{\lambda_1}) - g(w_{\lambda_2})) \le 0. \tag{D.1}$$

Hence $g(w(\alpha))$, and therefore $K(w(\alpha))$, is nondecreasing in $\alpha$. Uniqueness and compactness imply continuity, including at both endpoints.

The vector $w(0)$ uniquely minimizes $K$ over the floor-constrained simplex, establishing feasibility. If $w(1)$ is feasible, it minimizes $f$ over that entire simplex and is the handoff optimum. If the budget equals $K(w(0))$, strict convexity of $K$ makes $w(0)$ the only feasible vector. Otherwise, continuity provides $0 < \alpha^\star < 1$ at the budget boundary. Optimality for $f + \lambda g$, together with equality at that boundary, implies optimality for minimizing $f$ subject to the KL budget. Strict convexity of $f$ on the convex feasible set proves uniqueness.

In FloorNormalize, fixing an entry whose normalized mass is below its floor increases $Z_{\text{free}}$ for the remaining free entries. Previously fixed entries therefore remain at their floors. At least one entry is fixed in each nonterminal pass, and $\sum_G \ell_G < 1$ prevents all entries from being fixed below a valid remaining mass. For $K = |\Pi^+|$, direct scans take $O(K^2)$ time and $O(K)$ space per normalization. Bisection takes $O(\log(1/\xi))$ evaluations, giving $O(K^2 \log(1/\xi))$ time and $O(K)$ space. The products in Equation (14) can be evaluated in log space. Retaining the feasible endpoint makes every returned iterate satisfy the KL budget in exact arithmetic.

## D.2 Executed gain and oracle comparison

Equation (C.2) gives the execution-loss identity. Combining it with Proposition 2 proves the first part of Corollary 1. For the sequence of installed samplers, restore time indices. Algorithm 2 installs an adjustment only when $C_t$ evaluated at its actual weights is positive. Since $\hat{v}_t(G), v_t(G) \in [-M, M]$, $\beta_t(G) \ge 0$, and both weight vectors sum to one,

$$E_t(\Pi_t^+, w_t^{\text{exec}}) \le 4Mz_t \tag{D.2}$$

whenever both cuts consist of valid nodes. The negative part of the installed gain is therefore at most $4Mz_t$. On Keep it is zero, including windows in which the current cut is invalid. Summing and applying Proposition 1 proves Equation (16).

For an oracle comparison, fix a window with a valid current cut and again omit the time index. Let $(\Pi^{\text{or}}, w^{\text{or}})$ maximize $J$ among valid cuts with at most $m$ groups. Let $\Pi^{\text{exec}}$ be the installed cut, using $\Pi^-$ on Keep, and let $w^{\star,\text{exec}}$ be its unconstrained target weights under the conservative objective. Define

$$\Gamma = C^\star(\Pi^\star) - C^\star(\Pi^{\text{exec}}), \qquad R^{\text{exec}} = \eta^{-1} D_{\text{KL}}(w^{\text{exec}} \| w^{\star,\text{exec}}). \tag{D.3}$$

The cut-search gap $\Gamma$ is nonnegative and vanishes when the searched cut is installed. For every node, $-e(G) \le v(G) - L(G) \le 2\beta(G) + e(G)$. Adding and subtracting the lower-bound objectives for the oracle, searched, and installed policies gives

$$\begin{aligned} J(q^{\text{or}}) - J(q^{\text{exec}}) \le \Gamma + R^{\text{exec}} \\ + \sum_{G \in \Pi^{\text{or}}} w_G^{\text{or}} (2\beta(G) + e(G)) \\ + \sum_{G \in \Pi^{\text{exec}}} w_G^{\text{exec}} e(G). \end{aligned} \tag{D.4}$$

The dynamic program bounds the oracle's conservative objective by the searched objective value, and Equation (C.2) supplies the exact gap on the installed cut. On jointly covered windows, the two excess-error sums vanish. The remaining terms quantify uncertainty in the oracle's groups, selection of an executable cut, and displacement of its target weights.

## References

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